

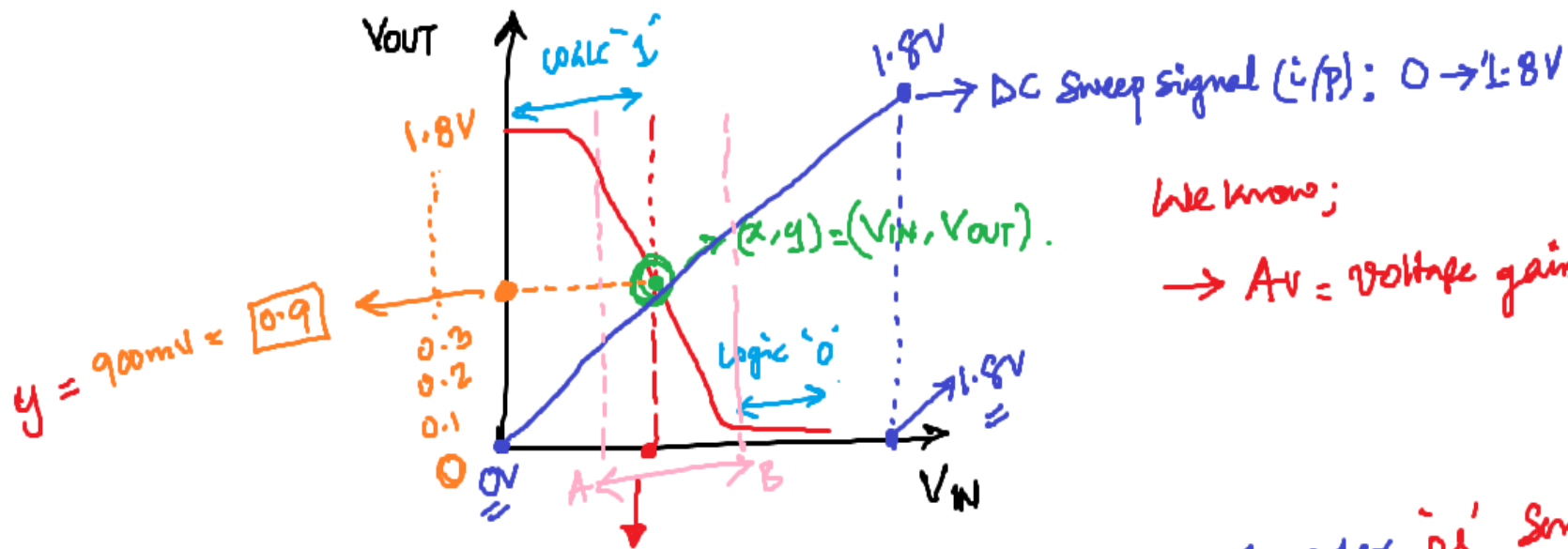
VTC OF INVERTER

LECTURE – 5 :: (08-04-2025)

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$y = 900\text{mV} = \boxed{0.9}$

We know;

$\rightarrow A_v = \text{voltage gain} = \frac{v_{out}}{v_{in}} = \frac{y}{x} =$

AB = transition period
where both the
MOSFETS are in
Saturation.

V_{TH} = Switching Threshold of CMOS inverter '01' Sometimes the "THRESHOLD VOLTAGE" = $\boxed{900\text{mV}}$
= x
 $\rightarrow$ eg: CMOS INVERTER.

NOTE

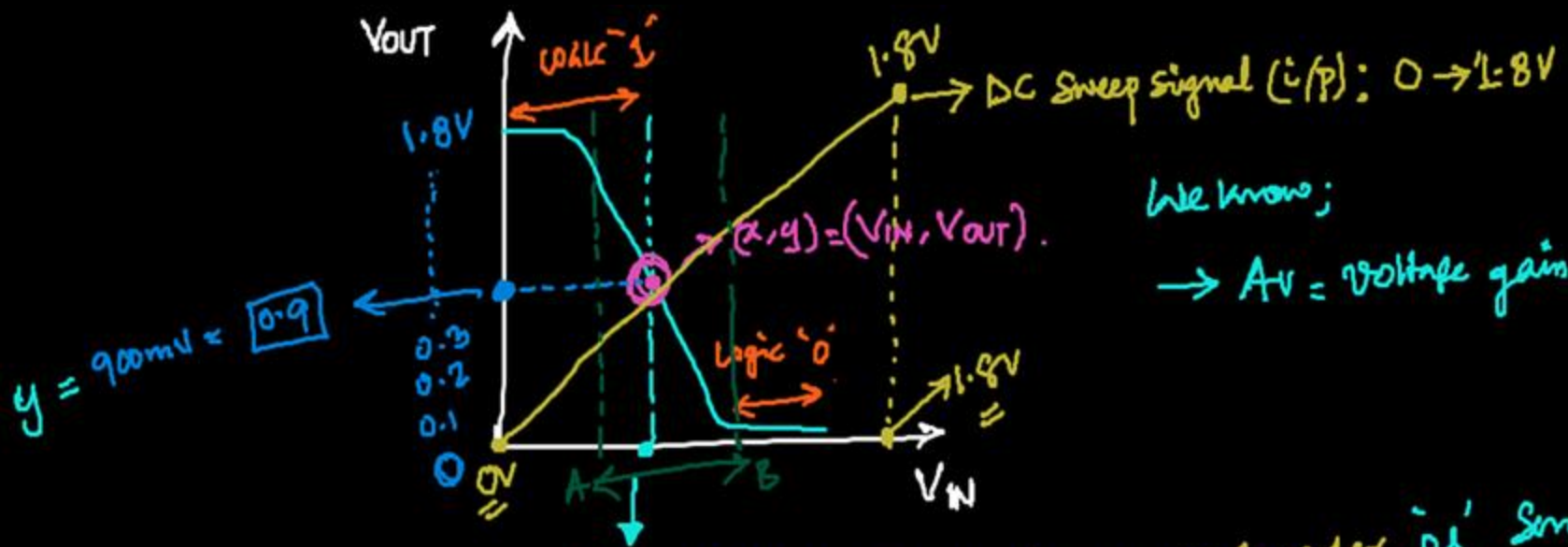
①

Switching Threshold is always of the CIRCUIT
it cannot be defined for any particular DEVICE $\rightarrow$ NMOS $\times$
 $\rightarrow$ PMOS $\times$

② VTC is always for a particular circuit. let say CMOS INVERTER $\checkmark$

VTC is Never for a NMOS/PMOS/OT-AMP/BJT/DIODE/RES/CAP $\times$.

③ DEVICES have OUTPUT/INPUT CHARACTERISTICS, whereas CIRCUITS have TRANSFER characteristics.



We know;

$$\rightarrow A_v = \text{voltage gain} = \frac{v_{out}}{v_{in}} = \frac{y}{x} =$$

$\overline{AB}$ = transition period
where both the
MOSFETS are in
Saturation.

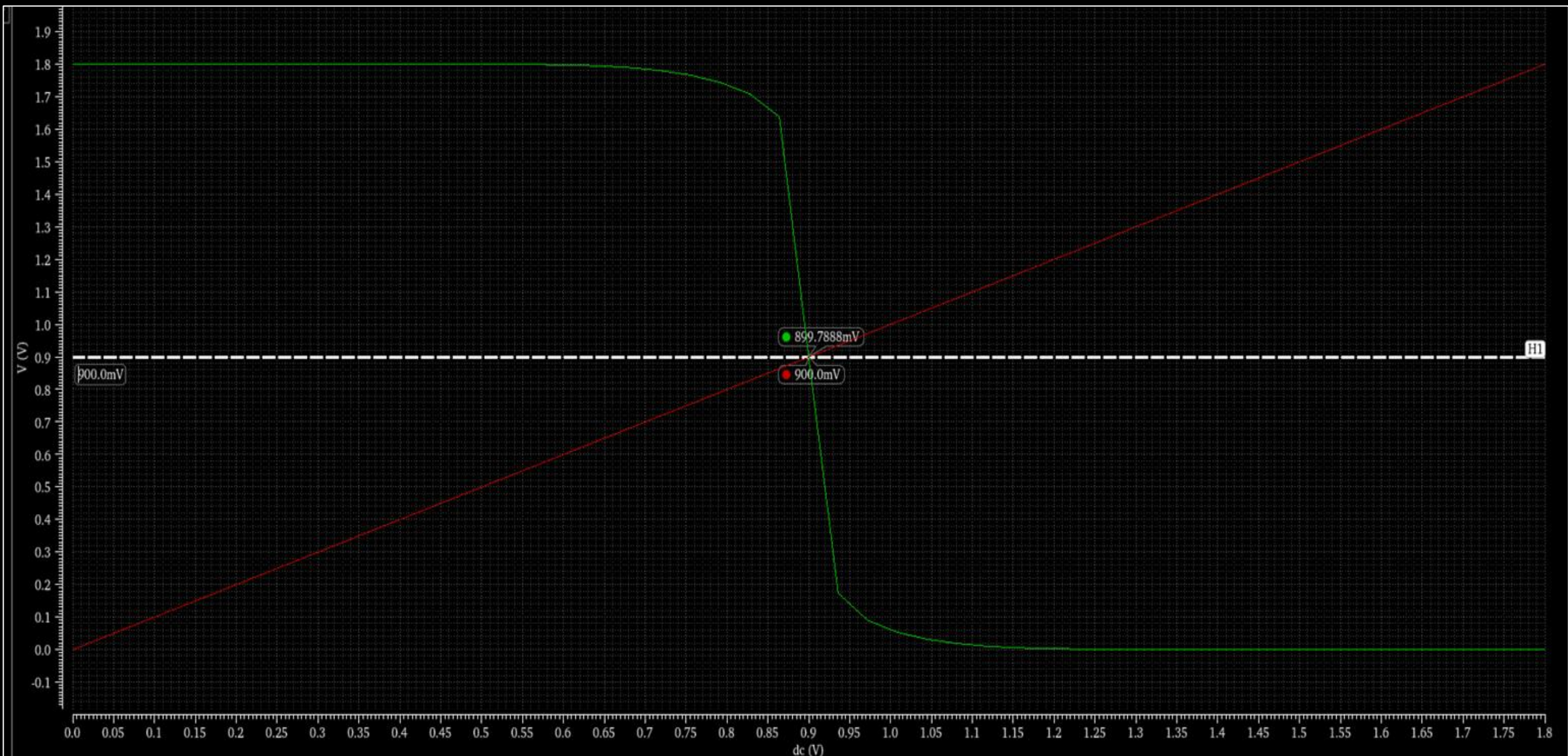
V_{TH} = Switching Threshold of CMOS inverter '01' Sometimes the "THRESHOLD VOLTAGE" = 900mV
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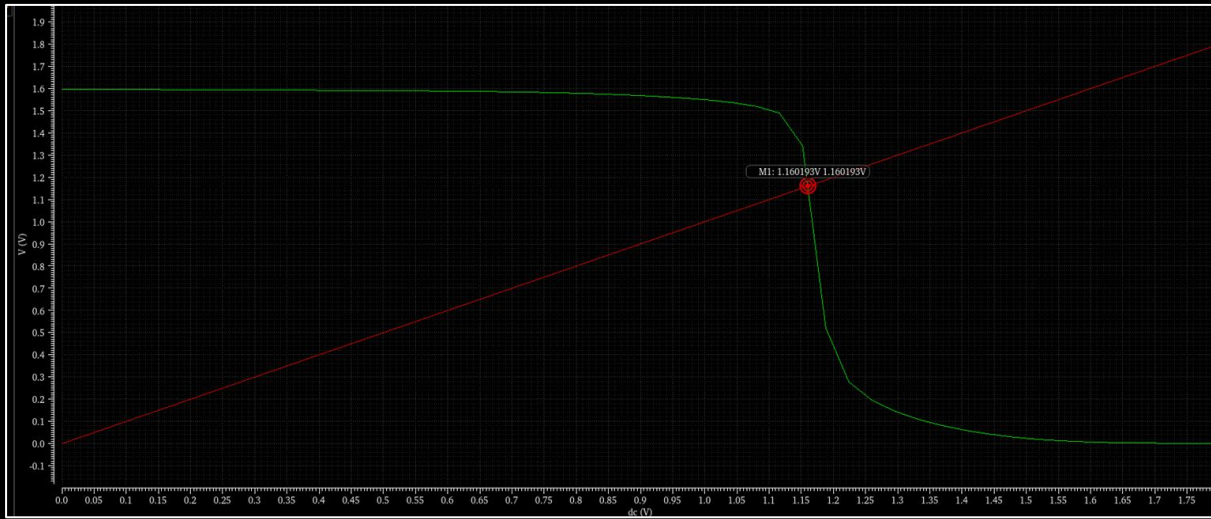
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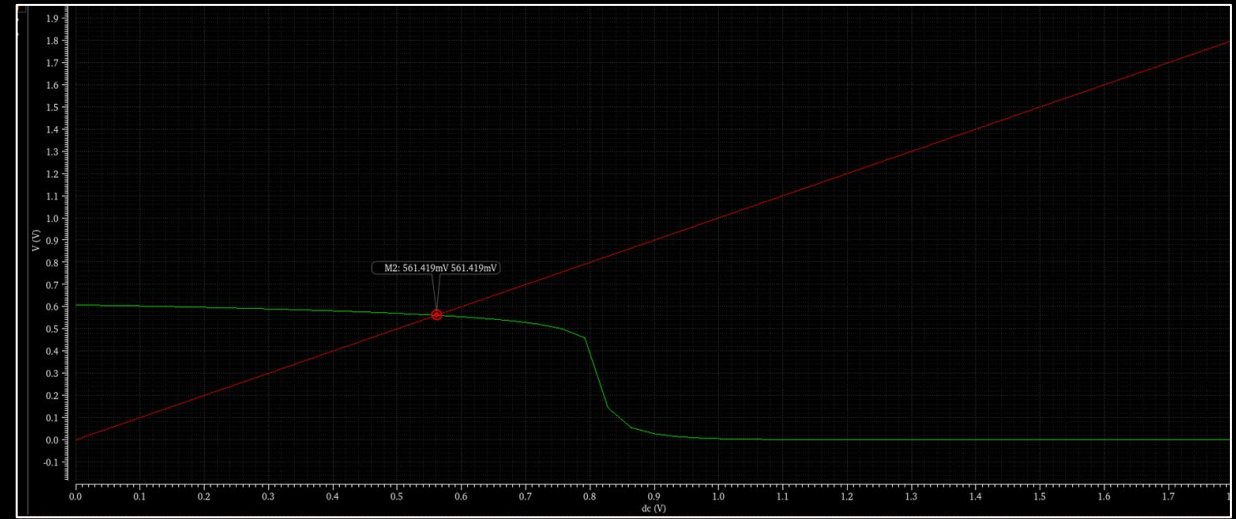
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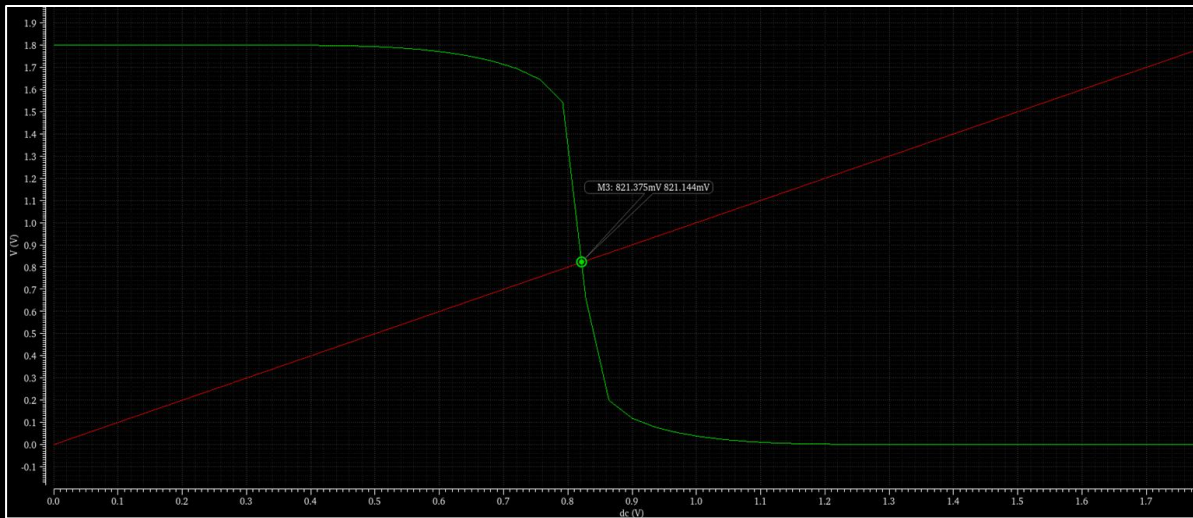
NO BODY BIAS



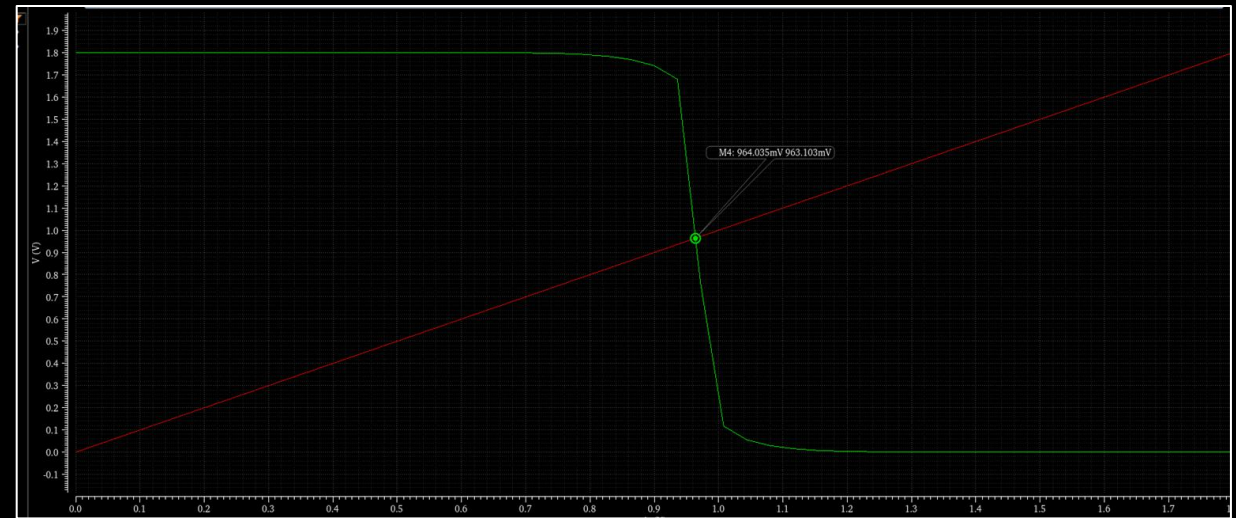
$V_B = +500\text{mV} \rightarrow \text{PMOS}$.



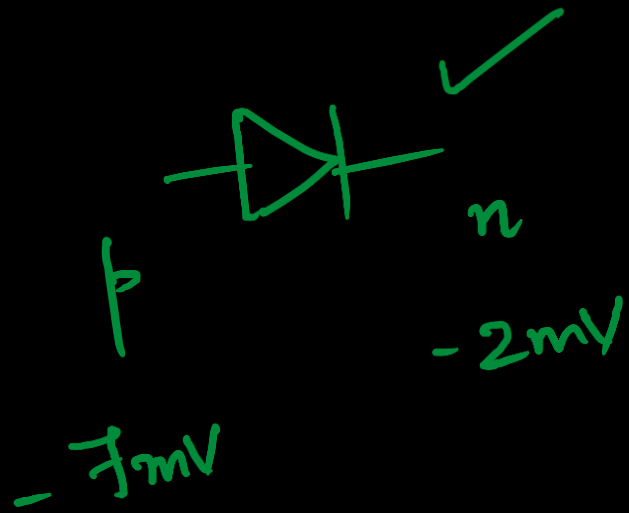
$V_B = -500\text{mV} \rightarrow \text{PMOS}$.



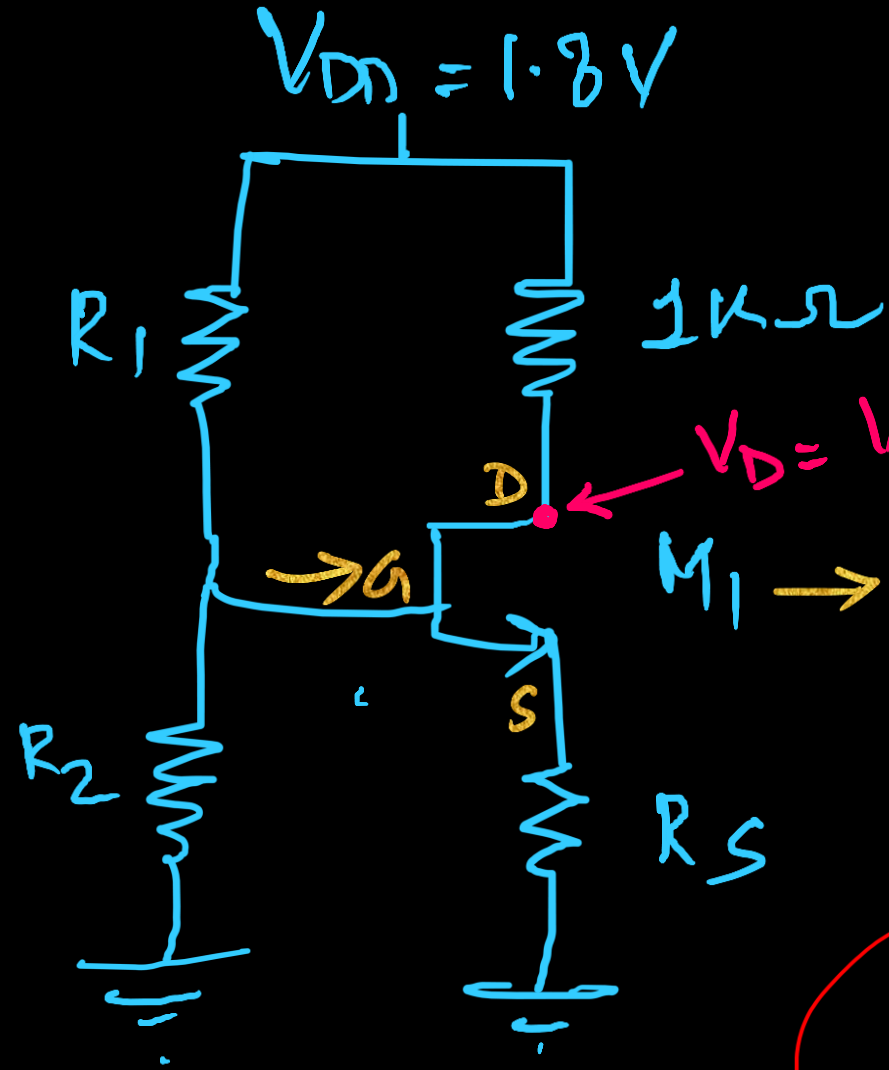
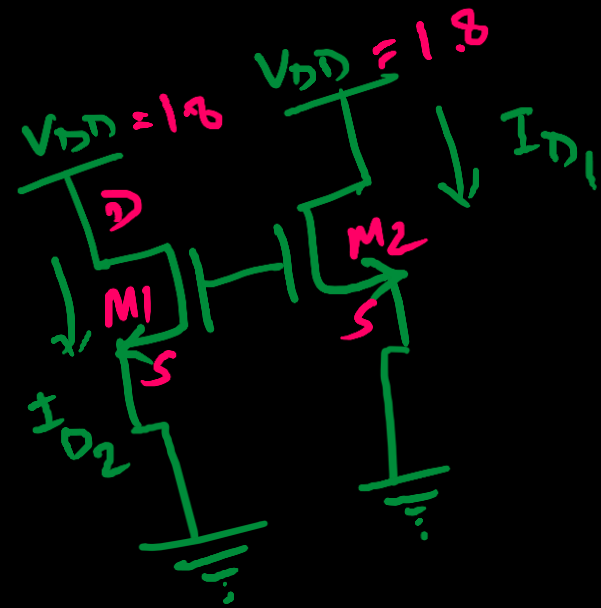
$V_B = +500\text{mV} \rightarrow \text{NMOS}$



$V_B = -500\text{mV} \rightarrow \text{NMOS}$.



$$R_{in} = \frac{R_1 \cdot R_2}{R_1 + R_2} \quad \checkmark$$



$$V_D = V_{DD} - I_D \cdot (1k\Omega) \quad \checkmark$$

$M_1 \rightarrow NMOS$

